



Linear Control Systems 25411

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Assignment 2

Fall 1404

Due Date: 1404/9/14

Note 1: Problems indicated in *purple* are bonus ones.

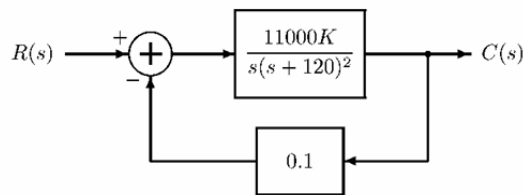
Note 2: You are expected to upload a *PDF* file that includes a *detailed derivation* procedure for each question. For the simulation part, along with the essential plots, it is necessary to *append* your *code* at the end of the *PDF* file. Additionally, provide the *source codes* in a separate folder, where the name of each source code corresponds to the number of the related question.

1 Non-ideal Operational Amplifier

An op amp is connected in an inverting configuration to provide an ideal gain of -10. The step response of the closed-loop system is approximately first order with 10 – 90% rise time of 2.2 μ s. Estimate the open-loop transfer function of the op amp.

2 Model Order Reduction and Response Characteristics

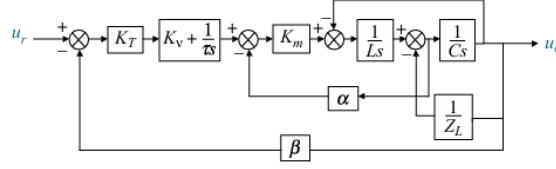
Consider a feedback system and for parts 1,2 and 3 assume $K = 110$.



1. Calculate the closed-loop transfer function. Find the location of the closed-loop poles. Is this system better approximated by a first-order system or a second-order system?
2. Based on the pole locations, find the approximation parameters.
3. Use matlab to plot the step response. From the plot find percent overshoot, peak time and rise time. From these measurements, calculate the approximation parameters.
4. Repeat all parts for $K = 2000$.

3 Dynamic Voltage Restorer

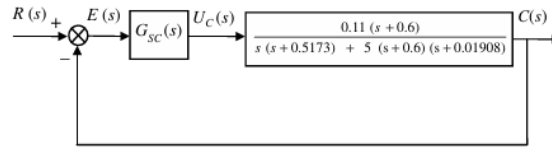
A dynamic voltage restorer(DVR) is a device that is connected in series to a power supply. It continuously monitors the voltage sags by applying the necessary extra voltage to maintain the load voltage constant. In the model shown in below, u_r represents the desired reference voltage, u_o is the output voltage, and Z_L is the load impedance. All other parameters are internal to the DVR.



1. Assuming $Z_L = \frac{1}{sC_L}$, and $\beta \neq 1$, find the type of system.
2. Find the steady-state error to a unit step input as function of β .

4 Hybrid Electric Vehicle

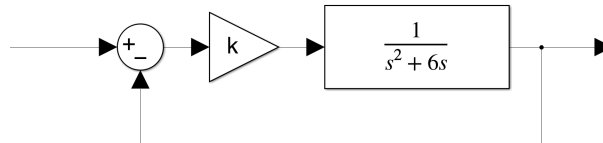
The Figure below shows the rearranged as unity feedback form of the speed control of an HEV.



1. Assume the speed controller is given as $G_{sc}(s) = K_{P_{SC}}$. Find the gain that yields a steady-state error $e_{step}(\text{inf}) = 1\%$.
2. Now assume that in order to reduce the steady-state error for step inputs, integration is added to the controller yielding $G_{sc}(s) = K_{P_{SC}} + \frac{K_{I_{SC}}}{s} = 100 + \frac{K_{I_{SC}}}{s}$. Find the value of integral gain that results in a steady-state error, $e_{ramp}(\text{inf}) = 2.5\%$.

5 Performance Optimization

Consider the closed-loop control system with unity feedback shown below. For each of the following optimization problems, determine the optimal value of the proportional controller gain k . The reference signal is the unit step, i.e. $r(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$.



1. minimize $J(k) = \text{settling time}$
2. minimize $J(k) = \text{settling time}$
subject to $\forall t : y(t) \leq r(t)$

3. minimize $J(k) = \int_0^\infty e^2(t) dt$

4. minimize $J(k) = \int_0^\infty (e^2(t) + u^2(t)) dt$
where $u(t)$ denotes the system input.

MATLAB Assignments

6 Symbolic e^{At}

Given the system transfer function $G(s)$ below

$$G(s) = \frac{1-s}{(s+1)(2s+1)}$$

1. Generate the impulse signal, then use the `lsim` function to find the impulse response of the system. (hint: you shouldn't use the `dirac` function)
2. Use `ilaplace` function to find the impulse response of the system directly from the given transfer function.
3. Find the state-space representation of $G(s)$ using `ss` function, then use the `expm` function to find the e^{At} matrix, then use B, C and e^{At} matrices to find the impulse response of the system from state-space representation of the system.
4. Use the `impz` function to find the impulse response of the system.
5. Plot and compare all of the impulse responses.
6. Use the `ilaplace` function and the $e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$ formula to find the e^{At} , then compare the result to the e^{At} matrix from problem 3.

Simulation Assignments

Note 3: Problems indicated in *Blue*, require screenshots or graphs.

7 System 1

State-space representation of the open-loop system is

$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = \frac{u(t) - kx_1(t) - x_2(t)}{m} \\ y(t) = x_1(t) \end{cases}$$

where $u(t)$ is the applied force to moving box and $x_1(t)$ is the position, $x_2(t)$ is the velocity and m is the mass of the moving box and k is the spring constant.

Assignment:

1. Is the described system LTI?
2. Write the state-space representation of the open-loop system using the standard controllable matrix form.
3. Find the open-loop transfer function $G(s) = \frac{Y(s)}{U(s)}$.
4. Determine $u(t)$ in the closed-loop system regarding proportional controller.
5. Write the state-space representation of the closed-loop system regarding proportional controller.
6. Find the closed-loop transfer function $T(s) = \frac{Y(s)}{R(s)}$ regarding proportional controller.
7. Given $m = 0.25$ and $k = 0.25$, determine the range of values for K_p that ensure closed-loop stability.
8. Using simulation, set the values of $m = 0.25$, $k = 0.25$ and $K_p = 200$ and observe the behavior of the closed-loop system. Explain whether the system is stable or not and provide a reason for your answer.
9. Implement the system in Simulink, and assign the values of $m = 0.25$, $k = 0.25$ and $K_p = 200$. Select the fixed-step size solver Euler (ode1) and set $T_s = 0.01$. Analyze the stability of the closed-loop system and justify your conclusion.
10. Modify the solver and the sampling-time T_s and perform the same simulation in Simulink. Is it possible to stabilize the closed-loop system by changing these parameters?
11. Determine the steady-state error of the system, then simulate different situations (change m , k and K_p) and compare the results to the first answer.
12. Given $m = 0.25$ and $k = 0.25$, calculate the value of K_p that results in a critically damped closed-loop system.
13. Assuming $m = 0.25$ and $k = 0.25$, determine the optimal value of K_p that minimizes the settling time of the system.
14. Find the condition on the parameters m and k that guarantees the existence of oscillations in the closed-loop system.

8 System 2

State-space representation of the open-loop system is

$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = \frac{2k(x_3(t) - x_1(t)) - x_2(t)}{m} \\ \dot{x}_3(t) = x_4(t) \\ \dot{x}_4(t) = \frac{u(t) - 2k(x_3(t) - x_1(t)) - x_4(t)}{M} \\ y_1(t) = x_1(t) \\ y_2(t) = x_3(t) \end{cases}$$

where $u(t)$ is the applied force to the moving frame and $x_1(t)$ is the position, $x_2(t)$ is the velocity and m is the mass of the moving box and $x_3(t)$ is the position, $x_4(t)$ is the velocity and M is the mass of the moving frame and k is the spring constant.

Assignment:

1. Write the state-space representation of the open-loop system using the standard matrix form.
2. Find the open-loop transfer functions $G_1(s) = \frac{Y_1(s)}{U(s)}$ and $G_2(s) = \frac{Y_2(s)}{U(s)}$.
3. Find the closed-loop transfer function $T_1(s) = \frac{Y_1(s)}{R(s)}$ while $u(t) = K_p(r(t) - y_1(t))$.
4. Find the closed-loop transfer function $T_2(s) = \frac{Y_2(s)}{R(s)}$ while $u(t) = K_p(r(t) - y_2(t))$.
5. Determine the steady-state error of the system, then simulate different situations (change m , k and K_p) and compare the results to the first answer.